



crop yield forecast

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The Problem and the Three Datasets

- The Problem: Agricultural planning and food security in Ethiopia require precise, localized yield and market revenue forecasts. Traditional estimations often lag behind harvest cycles, making it difficult for regional planners and farmers to optimize crop management, pricing, and resource allocation.
- Dataset 1 (Agronomic & Crop Records): Regional production logs containing crop types (e.g., Teff, Maize, Wheat, Barley), farm sizes, regional zones, and historical biological yields.
- Dataset 2 (Weather & Climate Data): Regional station records and multi-month environmental metrics focusing heavily on the Meher growing season rainfall, temperature, and altitude profiles.
- Dataset 3 (Socio-Economic & Management Inputs): Farm management indices tracking fertilizer quantities, certified improved seed usage, labor days invested, pest/disease flags, and proximity/distance to local markets.

Cleaning & Integration: Join Map & Key Numbers

- Data Cleaning: Handled missing values by replacing sentinel flags (such as -999) with proper NaN entries, standardized administrative region names, and harmonized categorical data types.
- Join Architecture: Integrated tabular farm records (Dataset 1) with regional meteorological matrices (Dataset 2) using spatial-temporal keys (Region $\times$ Crop Type $\times$ Meher Season), merged with input management metrics (Dataset 3).
- Key Numbers (from A4):
- Cleaned feature space encompassing 11 core predictive dimensions.
- Successfully harmonized multi-regional records covering major staple grains across key administrative zones (Amhara, Oromia, SNNPR, Tigray).

What the Data Says: Two Key Findings

- Finding 1 (Revenue Leaders): Teff commands the highest gross revenue per hectare (~151,715 ETB/ha) due to its high market price (~7,630 ETB/quintal), whereas Maize leads in biological output volume (~3.90 tons/ha).
- Finding 2 (Technology vs. Stress Impact): Utilizing certified improved seeds drives a consistent yield lift of 18% to 23%, whereas recorded pest and disease severity incidents introduce sharp biological penalties, driving yield losses of 26% to 29%.

Modeling & Evaluation

- Model Comparison: Evaluated Linear Regression, Gradient Boosting, and Random Forest Regressor. Random Forest emerged as the clear winner by capturing non-linear soil and weather interactions.
- Weather Ablation Study: Confirmed that incorporating regional weather features (Meher rainfall, altitude, soil quality) substantially improves model predictive performance, lowering error margins and elevating explanatory power.
- Cross-Validation & Tuning: Performed rigorous hyperparameter tuning (optimizing tree depth and estimator counts), landing a final test set RMSE of 0.5308 tons/ha and an R^2 score of 0.8589.

Error Analysis, Final Score & Next Steps

- Error Analysis: Residual plots indicate minor under-predictions on ultra-high-yield commercial outliers, primarily driven by unmeasured localized irrigation and micro-climate variations.
- Final Score: Test set achieved an outstanding R^2 of 0.8589 with an RMSE of 0.5308 tons/ha.
- Live Demo Link: <https://ethiopian-agri-forecast.streamlit.app> (Run live prediction during Q&A to showcase dynamic revenue calculation based on slider adjustments).
- What We'd Try Next: Incorporating dynamic price-elasticity modeling so market prices react automatically to total regional harvest supply shifts.